Assignment 9 Solution

Q.1. What does the term set represent in a linguistic variable?

(a) The universe of discourse

(b) The measure of the linguistic variable

(c) The collection of linguistic values for the variable

(d) None of the above

Solution: (c)

Q.2. Which of the following is true for syntactic rule (G)?

(a) It refers to the term set T(x) which can be characterized by a fuzzy set of a universe of discourse X.

(b) It refers to the way the linguistic values in the term set T(x) are generated.

(c) It specifies the procedure for computing the meaning of any linguistic value through specified membership functions.

(d) None of the above

Solution: (b)

Q.3. Which of the following is true for semantic rule (M)?

(a) It refers to the term set T(x) which can be characterized by a fuzzy set of a universe of discourse X.

(b) It refers to the way the linguistic values in the term set T(x) are generated.

(c) It specifies the procedure for computing the meaning of any linguistic value through specified membership functions.

(d) None of the above

Solution: (c)

Q.4. If a linguistic variable "Lazy" with the universe of discourse X = {1,2,3,4} is defined as,

$$\text{Lazy} = 1.0/1 + 0.5/2 + 0.3/3 + 0.2/4$$

What will be the value of "Very Lazy"?

(a) 1/1 + 0.250/2 + 0.090/3 + 0.040/4

(b) 1/1 + 0.125/2 + 0.027/3 + 0.064/4

(c) 1/1 + 0.707/2 + 0.547/3 + 0.447/4

(d) 1/1 + 0.500/2 + 0.700/3 + 0.800/4

(e) None of the above

Solution: (a)

The value of "Very Lazy" will be:

$$\sum_{x \in X} \mu_{Very\ Lazy}(x)/x = \sum_{x \in X} \left[\mu_{Lazy}(x)\right]^2/x$$

$$\sum_{x \in X} \mu_{Very\ Lazy}(x)/x = (1.0)^2/1 + (0.5)^2/2 + (0.3)^2/3 + (0.2)^2/4$$

$$\sum_{x \in X} \mu_{Very\ Lazy}(x)/x = 1.000/1 + 0.250/2 + 0.090/3 + 0.040/4$$

Q.5. If a linguistic variable "Lazy" with the universe of discourse X = {1,2,3,4} is defined as,

Lazy = 1.0/1 + 0.5/2 + 0.3/3 + 0.2/4

What will be the value of "More Or LessLazy"?

(a) 1/1 + 0.250/2 + 0.090/3 + 0.040/4

(b) 1/1 + 0.125/2 + 0.027/3 + 0.064/4

(c) 1/1 + 0.707/2 + 0.547/3 + 0.447/4

(d) 1/1 + 0.500/2 + 0.700/3 + 0.800/4

(e) None of the above

Solution: (c)

The value of "More Or Less Lazy" will be:

$$\sum_{x \in X} \mu_{More\ Or\ Less\ Lazy}(x)/x = \sum_{x \in X} \left[\mu_{Lazy}\right]^{1/2}/x$$

$$\sum_{x \in X} \mu_{More\ Or\ Less\ Lazy}(x)/x = (1.0)^{0.5}/1 + (0.5)^{0.5}/2 + (0.3)^{0.5}/3 + (0.2)^{0.5}/4$$

$$\sum_{x \in X} \mu_{More\ Or\ Less\ Lazy}(x)/x = 1.0/1 + 0.707/2 + 0.547/3 + 0.447/4$$

Q.6. If a linguistic variable "Lazy" with the universe of discourse X = {1,2,3,4} is defined as,

Lazy = 1.0/1 + 0.5/2 + 0.3/3 + 0.2/4

What will be the value of "Very VeryLazy"?

(a) 1/1 + 0.250/2 + 0.090/3 + 0.040/4

(b) 1/1 + 0.125/2 + 0.027/3 + 0.064/4

(c) 1/1 + 0.707/2 + 0.547/3 + 0.447/4

(d) 1/1 + 0.500/2 + 0.700/3 + 0.800/4

(e) None of the above

Solution: (e)

The value of "Very VeryLazy" will be:

$$\sum_{x \in X} \mu_{Very\ Very\ Lazy}(x)/x = \sum_{x \in X} \left[\mu_{Very}\left(\mu_{Very\ Lazy}(x)\right)\right]^2/x = \sum_{x \in X} \left[\mu_{Lazy}(x)\right]^4/x$$

$$\sum_{x \in X} \mu_{Very\ Very\ Lazy}(x)/x = (1.0)^4/1 + (0.5)^4/2 + (0.3)^4/3 + (0.2)^4/4$$

$$\sum_{x \in X} \mu_{Very\ Very\ Lazy}(x)/x = 1.0/1 + 0.0625/2 + 0.0081/3 + 0.0016/4$$

Q.7. If a linguistic variable "Lazy" with the universe of discourse X = {1,2,3,4} is defined as,

Lazy = 1.0/1 + 0.5/2 + 0.3/3 + 0.2/4

What will be the value of "NOT(Lazy)"?

(a) 1/1 + 0.250/2 + 0.090/3 + 0.040/4

(b) 1/1 + 0.125/2 + 0.027/3 + 0.064/4

(c) 1/1 + 0.707/2 + 0.547/3 + 0.447/4

(d) 1/1 + 0.500/2 + 0.700/3 + 0.800/4

(e) None of the above

Solution: (e)

The value of "NOT(Lazy)" is defined as,

$$NOT(Lazy) = \neg Lazy = \int_{x \in X} \left[1 - \mu_{Lazy}(x)\right]/x$$

$$NOT(Lazy) = (1 - 1.0)/1 + (1 - 0.5)/2 + (1 - 0.3)/3 + (1 - 0.2)/4$$

$$NOT(Lazy) = 0/1 + 0.5/2 + 0.7/3 + 0.8/4$$

Q.8. If fuzzy sets A and B with the universe of discourse X are defined as,

A = 1.0/1 + 0.5/2 + 0.3/3 + 0.2/4

B = 0.5/1 + 1.0/2 + 0.7/3 + 0.1/4

Find A AND B?

(a) 0.0/1 + 0.5/2 + 0.7/3 + 0.8/4

(b) 0.5/1 + 0.5/2 + 0.3/3 + 0.1/4

(c) 1.0/1 + 1.0/2 + 0.7/3 + 0.2/4

(d) None of the above

Solution: (b)

$$A\ AND\ B = \min(1.0,0.5)/1 + \min(0.5,1.0)/2 + \min(0.3,0.7)/3 + \min(0.2,0.1)/4$$

$$A\ AND\ B = 0.5/1 + 0.5/2 + 0.3/3 + 0.1/4$$

Q.9. If fuzzy sets A and B with the universe of discourse X are defined as,

A = 1.0/1 + 0.5/2 + 0.3/3 + 0.2/4

B = 0.5/1 + 1.0/2 + 0.7/3 + 0.1/4

Find A OR B?

(a) 0.0/1 + 0.5/2 + 0.7/3 + 0.8/4

(b) 0.5/1 + 0.5/2 + 0.3/3 + 0.1/4

(c) 1.0/1 + 1.0/2 + 0.7/3 + 0.2/4

(d) None of the above

Solution: (c)

$$A\ OR\ B = \max(1.0,0.5)/1 + \max(0.5,1.0)/2 + \max(0.3,0.7)/3 + \max(0.2,0.1)/4$$

$$A\ OR\ B = 1.0/1 + 1.0/2 + 0.7/3 + 0.2/4$$

Q.10. How can the composite linguistic term "Slightly Light" be obtained from the fuzzy set for "Light"?

(a) By concentrating the fuzzy set

(b) By taking the intersection of fuzzy sets

(c) By dilating the fuzzy set

(d) By applying the complement operation to the fuzzy set

(e) None of the above

Solution: (c)